

KUNAL TOMAR

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Summary

B.Tech (CSE, AI specialization) graduate from Medicaps University, with internship experience at the Applied Cognitive Science (ACS) Lab, IIT Mandi and the Centre for Intelligent Robotics (CIR), IIIT Allahabad. Co-author of a paper accepted at PETRA 2026; built end-to-end ML pipelines in Python with careful, leakage-controlled evaluation, and contributed to VR research simulations in Unity/C#. Experienced across the research pipeline, from co-developing study apparatus through data collection to analysis. Seeking roles in machine learning, deep learning, applied research, and software engineering.

Education

Medicaps University

August 2022 - May 2026

B.Tech in Computer Science & Engineering, Specialization in AI | CGPA: 8.89/10

Indore, MP

Publications

- K. Chand, A. Mehra, S. Sonam, **K. Tomar**, Y. K. Meena, A. Kumar, G. Kaur, A. K. Rao, R. Govindaraji, A. Bhavsar, V. Dutt. “Asymmetric Behavioral Adaptation Following Initial Difficulty Exposure in a VR-Based Shooting Task.” Accepted at **PETRA 2026** (19th Int. Conf. on Pervasive Technologies Related to Assistive Environments), Crete, Greece, July 2026.

Relevant Coursework

Machine Learning (CS3EL15), Artificial Intelligence (CS3EA10), Natural Language Processing (CS3EA06), Design & Analysis of Algorithms (CS3CO42), Data Structures (CS3CO31), Statistical Analysis (CS3EL11), Exploratory Data Analytics (OE00075), Mobile Virtual Reality & Artificial Intelligence (NPTEL, IIT Mandi - noc25-cs80).

Experience

Indian Institute of Technology Mandi

January 2026 - April 2026

Research Intern, ACS Lab | NPTEL Winter Internship 2025 (WIN25277R)

Mandi, HP

- Co-authored an 11-author paper accepted at PETRA 2026 (see Publications) following a 40-participant 2×2 mixed-factorial VR shooting study under the DRDO-LSRB “Adaptive Cognitive Training for Indian Defence” project, in collaboration with DIPR, DRDO.
- Co-developed and hardened a Unity 6 VR base-camp defence simulator (OpenXR, C#, HTC Vive Pro) for the same DRDO study: tactical enemy AI (perception, cover, squad coordination), runtime session-selection UI, and event/research logging across all 80 sessions of participant data collection.
- For a separate research study, designed a leakage-controlled ML evaluation framework (per-fold preprocessing, participant-level StratifiedGroupKFold, dummy baselines, permutation testing) on a multimodal (behavioural + HRV + EEG) VR Landslide dataset (20 participants × 4 conditions); documented and corrected two coupled inflation modes that had previously yielded a misleading $R^2 \approx 0.84$.
- Co-developed a Unity URP VR landslide simulator (OpenXR, C#, Meta Quest 3) under the Tata Trusts SAFER Hills programme with two locomotion scenarios, driving (Thrustmaster racing wheel) and walking (XR Interaction Toolkit, joystick locomotion); built a pre-computed replay system that ingests CSV outputs from r.avafLOW / RAMMS / DualSPHysics and renders thousands of debris instances via GPU instancing; and added rock object pooling, downhill-velocity physics, VFX Graph debris/dust, road-barrier collider fixes, and a custom Unity Editor tool (*Debris Flow Converter*).
- Ran end-to-end data collection for the SAFER Hills driving scenario with 35 participants: informed consent, demographic and prior-experience questionnaires, baseline EEG and HRV/PPG recording, familiarisation trials, in-session capture, post-study survey, and structured data organisation.
- Mentored by Prof. (Dr.) Varun Dutt; signed Internship Certificate available [here](#).

Indian Institute of Information Technology Allahabad

June 2025 - August 2025

AI & Robotics Summer Intern, Centre for Intelligent Robotics (CIR)

Prayagraj, UP

- Co-developed and validated an autonomous UAV navigation system for GPS-denied environments, implementing real-time obstacle detection and path planning; achieved <1 m navigation accuracy in validation tests.
- Integrated PX4 Autopilot with ROS2 using Offboard Mode to enable 10 Hz trajectory control and seamless communication between planning, perception, and control nodes.
- Co-engineered sensor-noise filtering and safety-override mechanisms using OpenCV and C++, reducing false positives and ensuring <100 ms emergency response.
- Applied Python, C++, ROS2, PX4, Gazebo, TensorFlow, PyTorch, and Git to design, test, and optimise AI-driven robotic systems at CIR.
- Mentored by Dr. Surya Prakash (Programme Chair), under Prof. G C Nandi (Head, CIR); signed Summer Training Certificate available [here](#).

Projects

Autonomous Multi-Robot Coordination System with Real-Time Computer Vision and ROS2 Integration

November 2025

Python, C++, ROS2, OpenCV, PyBullet

- Designed an autonomous multi-robot coordination system integrating ROS2 communication, real-time computer vision, and inverse kinematics for synchronised control of dual 7-DOF robotic arms in a PyBullet simulation.
- Implemented a multi-camera vision pipeline using OpenCV, Hough transforms, and Bayesian confidence scoring for object detection, environment perception, and real-time spatial awareness.
- Built a ROS2 communication architecture with custom message definitions and tuned Quality-of-Service (QoS) settings for reliable, synchronised multi-robot operation.
- Engineered a damped least squares inverse kinematics solver with a collision-avoidance framework, enabling smooth trajectory planning, precise end-effector control, and safe cooperative motion.

OptiConnect: Connect-4 AI with Non-Blocking Web UI

December 2024

JavaScript, HTML5, CSS3, Web Workers — [Live Demo](#)

- Built a browser-based 7×7 Connect-4 game with an AI opponent using JavaScript and Web Workers, keeping the UI responsive while the worker thread evaluated moves via a depth-limited minimax algorithm with randomised tie-breaking.
- Designed a modular architecture separating UI handling and background computation through postMessage/onmessage communication, structured to support future alpha-beta pruning and heuristic evaluation.

Technical Skills

Languages: Python, C, C++, R, SQL

Robotics: ROS2, PX4, Gazebo, PyBullet

VR & Simulation: Unity (C#, OpenXR, URP, XR Interaction Toolkit, Shader Graph, VFX Graph)

ML & Data: scikit-learn, pandas, NumPy, SciPy, OpenCV, Matplotlib, statsmodels

Physiological Sensors: EmotiBit (PPG/HRV), Muse S Athena (EEG)

Environments & Tools: Git, GitHub, Jupyter, Google Colab, Anaconda, RStudio, VS Code, VirtualBox, Linux (Ubuntu), Windows

Awards, Certifications & Leadership

- Elite + Gold Certificate, Top 1% (consolidated score 93%) in *Mobile Virtual Reality and Artificial Intelligence*, NPTEL / IIT Mandi (Aug-Sep 2025) - [\[Certificate\]](#).
- Selected for the NPTEL Winter Internship 2025 cohort (Internship ID: WIN25277R), hosted by Prof. Varun Dutt at IIT Mandi for 17 weeks.
- Led a 3-member team “Hidden Layers” to a top 4% finish (Rank 3,270 of 82,790 teams) at the Amazon ML Challenge 2025.
- Rank 19 (top 0.5%) in CodeKaze 2023 (Coding Ninjas, Round 2), a national competitive programming contest.

References

Prof. (Dr.) Varun Dutt - Principal Investigator, Applied Cognitive Science (ACS) Lab; faculty, School of Computing & Electrical Engineering, IIT Mandi. Email: varun@iitmandi.ac.in. *Letter of Recommendation (dated 29 April 2026) available upon request.*